

Constrained Update for Continual Reinforcement Learning

Setup

Tasks $1, \dots, k$ have appeared; k is the current task and $i \in \{1, \dots, k\}$ a generic index. Task i is the MDP

$$M_i = (S, A, P_i, r_i, \rho_i, \gamma), \quad (1)$$

with states S , actions A , transition law $P_i(s' | s, a)$, reward $r_i(s, a)$, initial distribution ρ_i , and discount $\gamma \in (0, 1)$.

The local policy is $\pi_L = \pi_\theta$ and the global policy is $\pi_G = \pi_\phi$, with parameters θ and ϕ . The two are trained in alternation: during a local update, θ is optimized while π_G is held fixed; during a global update, ϕ is optimized while π_L is held fixed. Each local update starts from the global,

$$\theta^{(0)} = \phi. \quad (2)$$

We use uniform task weights for now, $\omega_i = 1/k$.

A length- T trajectory is $\tau = (s_0, a_0, \dots, s_T, a_T, s_{T+1})$, with law under policy π on task i

$$p_i^\pi(\tau) = \rho_i(s_0) \prod_{t=0}^T \pi(a_t | s_t) P_i(s_{t+1} | s_t, a_t). \quad (3)$$

The discounted return and the value of π on task i are

$$G_i(\tau) = \sum_{t=0}^T \gamma^t r_i(s_t, a_t), \quad V_i^\pi = \mathbb{E}_{\tau \sim p_i^\pi} [G_i(\tau)]. \quad (4)$$

We assume access to all previously seen environments. Thus old-task terms are estimated from fresh rollouts in old environments rather than from a stored replay buffer; the method is replay-free in that sense.

The objective and the constraint

We score a policy by the weighted sum of its per-task values, the weights $\omega_i \geq 0$ fixing how much each task counts,

$$J_k(\pi) = \sum_{i=1}^k \omega_i V_i^\pi. \quad (5)$$

For a new policy π_{new} and a reference policy π_{ref} , the total improvement of one policy over another can be split into current-task improvement and past-task improvement:

$$\begin{aligned} J_k(\pi_{\text{new}}) - J_k(\pi_{\text{ref}}) &= \sum_{i=1}^k \omega_i V_i^{\pi_{\text{new}}} - \sum_{i=1}^k \omega_i V_i^{\pi_{\text{ref}}} \\ &= \sum_{i=1}^k \omega_i (V_i^{\pi_{\text{new}}} - V_i^{\pi_{\text{ref}}}) \\ &= \underbrace{\omega_k (V_k^{\pi_{\text{new}}} - V_k^{\pi_{\text{ref}}})}_{\text{current task}} + \underbrace{\sum_{i=1}^{k-1} \omega_i (V_i^{\pi_{\text{new}}} - V_i^{\pi_{\text{ref}}})}_{\text{past tasks}}. \end{aligned} \quad (6)$$

The local past-task constraints are per-task, because an aggregate past-task constraint can hide failure on one old task. On a number line, the global value is the fixed reference during the local update:

$$\text{lower value} \quad \longleftarrow \quad V_i^{\pi_L} < V_i^{\pi_G} \text{ (penalty)} \quad \Big| \quad V_i^{\pi_G} \quad \Big| \quad V_i^{\pi_L} \geq V_i^{\pi_G} \text{ (no penalty)} \quad \longrightarrow \quad \text{higher value}.$$

A plain square would increase on both sides of the reference. We only penalize the side where the local value is below the global reference, because being above the reference is not forgetting.

Local objective

The local model updates π_L with the global held fixed. It maximizes its current-task lead over the global and is kept within per-task tolerances $\epsilon_i \geq 0$ on the past tasks. With the past shortfall

$$F_{L,i}(\pi_L) = \begin{cases} (V_i^{\pi_G} - V_i^{\pi_L})^2, & \text{if } V_i^{\pi_L} < V_i^{\pi_G}, \\ 0, & \text{if } V_i^{\pi_L} \geq V_i^{\pi_G}, \end{cases} \quad i = 1, \dots, k-1, \quad (7)$$

the local problem, its Lagrangian, and its saddle point are

$$\max_{\pi_L} \omega_k(V_k^{\pi_L} - V_k^{\pi_G}) \quad \text{s.t.} \quad F_{L,i}(\pi_L) \leq \epsilon_i, \quad i = 1, \dots, k-1, \quad (8)$$

$$\mathcal{L}_L(\pi_L, \lambda) = \omega_k(V_k^{\pi_L} - V_k^{\pi_G}) - \sum_{i=1}^{k-1} \lambda_i (F_{L,i}(\pi_L) - \epsilon_i), \quad (9)$$

$$\max_{\pi_L} \min_{\lambda_i \geq 0} \mathcal{L}_L(\pi_L, \lambda). \quad (10)$$

Global objective

The global model updates π_G with the local held fixed. It maximizes its past lead over the local and is kept within tolerance $\epsilon \geq 0$ of the local on the current task. The same one-sided idea is used here: the global model is penalized only when its current-task value is below the local model's current-task value. With the current-task shortfall

$$F_G(\pi_G) = \begin{cases} (V_k^{\pi_L} - V_k^{\pi_G})^2, & \text{if } V_k^{\pi_G} < V_k^{\pi_L}, \\ 0, & \text{if } V_k^{\pi_G} \geq V_k^{\pi_L}, \end{cases} \quad (11)$$

the global problem, its Lagrangian, and its saddle point are

$$\max_{\pi_G} \sum_{i=1}^{k-1} \omega_i(V_i^{\pi_G} - V_i^{\pi_L}) \quad \text{s.t.} \quad F_G(\pi_G) \leq \epsilon, \quad (12)$$

$$\mathcal{L}_G(\pi_G, \mu) = \sum_{i=1}^{k-1} \omega_i(V_i^{\pi_G} - V_i^{\pi_L}) - \mu (F_G(\pi_G) - \epsilon), \quad (13)$$

$$\max_{\pi_G} \min_{\mu \geq 0} \mathcal{L}_G(\pi_G, \mu). \quad (14)$$

The two are opposite term by term. In alternation the current gap $V_k^{\pi_L} - V_k^{\pi_G}$ is raised by the local objective and lowered by the global current-task constraint, while each past gap $V_i^{\pi_G} - V_i^{\pi_L}$ is raised by the global objective and lowered by the local constraint.

Notes for experiments. A scalar value constraint may not always imply that the global policy has absorbed the local policy's behavior; this should be checked empirically before adding a distillation term. We should also monitor feasibility: persistent constraint violations or diverging multipliers indicate task conflict or insufficient model capacity. Since the old-task reference is the current global value, cumulative forgetting should be checked in final per-task evaluation.

Local update

Optimizing over the local policy means optimizing over its parameters θ , since $\pi_L = \pi_\theta$. A saddle point of (10) is approached by ascending $\mathcal{L}_L$ in θ and descending it in λ . Both directions need $\nabla_\theta \mathcal{L}_L$, built from the per-task gradients $\nabla_\theta V_i^{\pi_\theta}$, so derive that for an arbitrary task i .

The chain rule on the logarithm gives $\nabla_\theta \log p_i^{\pi_\theta}(\tau) = \nabla_\theta p_i^{\pi_\theta}(\tau) / p_i^{\pi_\theta}(\tau)$, and multiplying both sides by $p_i^{\pi_\theta}(\tau)$,

$$\nabla_\theta p_i^{\pi_\theta}(\tau) = p_i^{\pi_\theta}(\tau) \nabla_\theta \log p_i^{\pi_\theta}(\tau).$$

With finite S and A the value (4) is a sum over all trajectories, $V_i^{\pi_\theta} = \sum_\tau p_i^{\pi_\theta}(\tau) G_i(\tau)$. Substituting the identity,

$$\begin{aligned}\nabla_\theta V_i^{\pi_\theta} &= \nabla_\theta \sum_\tau p_i^{\pi_\theta}(\tau) G_i(\tau) \\ &= \sum_\tau \nabla_\theta p_i^{\pi_\theta}(\tau) G_i(\tau) \\ &= \sum_\tau p_i^{\pi_\theta}(\tau) \nabla_\theta \log p_i^{\pi_\theta}(\tau) G_i(\tau) \\ &= \mathbb{E}_{\tau \sim p_i^{\pi_\theta}} [G_i(\tau) \nabla_\theta \log p_i^{\pi_\theta}(\tau)].\end{aligned}\tag{15}$$

In the trajectory law (3) only the policy factors depend on θ , so ρ_i and P_i drop from the log-gradient,

$$\begin{aligned}\nabla_\theta \log p_i^{\pi_\theta}(\tau) &= \nabla_\theta \left[\log \rho_i(s_0) + \sum_{t=0}^T \log \pi_\theta(a_t | s_t) + \sum_{t=0}^T \log P_i(s_{t+1} | s_t, a_t) \right] \\ &= \underbrace{\nabla_\theta \log \rho_i(s_0)}_0 + \sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) + \underbrace{\sum_{t=0}^T \nabla_\theta \log P_i(s_{t+1} | s_t, a_t)}_0 \\ &= \sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t).\end{aligned}\tag{16}$$

Substituting (16) into (15) and factoring out the return, which does not carry the index t , gives the per-task policy gradient

$$\nabla_\theta V_i^{\pi_\theta} = \mathbb{E}_{\tau \sim p_i^{\pi_\theta}} \left[G_i(\tau) \sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t | s_t) \right].\tag{17}$$

During the local update, the global values $V_k^{\pi_G}$ and $V_i^{\pi_G}$ are fixed with respect to θ . For the squared shortfall in (7),

$$\nabla_\theta F_{L,i}(\pi_\theta) = \begin{cases} -2(V_i^{\pi_G} - V_i^{\pi_\theta}) \nabla_\theta V_i^{\pi_\theta}, & \text{if } V_i^{\pi_\theta} < V_i^{\pi_G}, \\ 0, & \text{if } V_i^{\pi_\theta} \geq V_i^{\pi_G}. \end{cases}\tag{18}$$

Differentiating (9) gives

$$\nabla_\theta \mathcal{L}_L(\pi_\theta, \lambda) = \omega_k \nabla_\theta V_k^{\pi_\theta} + \sum_{i=1}^{k-1} \lambda_i \begin{cases} 2(V_i^{\pi_G} - V_i^{\pi_\theta}) \nabla_\theta V_i^{\pi_\theta}, & \text{if } V_i^{\pi_\theta} < V_i^{\pi_G}, \\ 0, & \text{if } V_i^{\pi_\theta} \geq V_i^{\pi_G}, \end{cases}\tag{19}$$

each per-task gradient given by (17).

Drawing N trajectories $\tau^{(1)}, \dots, \tau^{(N)}$ on task i under π_θ , the per-task value and gradient are estimated by

$$\hat{V}_i^{\pi_\theta} = \frac{1}{N} \sum_{n=1}^N G_i(\tau^{(n)}),\tag{20}$$

and

$$\overline{\nabla_\theta V_i^{\pi_\theta}} = \frac{1}{N} \sum_{n=1}^N G_i(\tau^{(n)}) \sum_{t=0}^T \nabla_\theta \log \pi_\theta(a_t^{(n)} | s_t^{(n)}).\tag{21}$$

During the same local update, $\hat{V}_i^{\pi_G}$ denotes the rollout estimate of the held-fixed global value on task i . The local primal update is

$$\theta^{(m+1)} = \theta^{(m)} + \alpha \left[\omega_k \overline{\nabla_\theta V_k^{\pi_\theta}} + \sum_{i=1}^{k-1} \lambda_i^{(m)} \begin{cases} 2(\hat{V}_i^{\pi_G} - \hat{V}_i^{\pi_\theta}) \overline{\nabla_\theta V_i^{\pi_\theta}}, & \text{if } \hat{V}_i^{\pi_\theta} < \hat{V}_i^{\pi_G}, \\ 0, & \text{if } \hat{V}_i^{\pi_\theta} \geq \hat{V}_i^{\pi_G} \end{cases} \right].\tag{22}$$

The local multiplier update is

$$\lambda_i^{(m+1)} = \max \left\{ 0, \lambda_i^{(m)} + \eta_\lambda (\hat{F}_{L,i}^{(m)} - \epsilon_i) \right\}, \quad i = 1, \dots, k-1,\tag{23}$$

where

$$\hat{F}_{L,i}^{(m)} = \begin{cases} (\hat{V}_i^{\pi_G} - \hat{V}_i^{\pi_{\theta^{(m)}}})^2, & \text{if } \hat{V}_i^{\pi_{\theta^{(m)}}} < \hat{V}_i^{\pi_G}, \\ 0, & \text{if } \hat{V}_i^{\pi_{\theta^{(m)}}} \geq \hat{V}_i^{\pi_G}. \end{cases}\tag{24}$$

This increases the multiplier when a past-task constraint is violated and decreases it otherwise, while keeping the multiplier nonnegative.

Global update

Optimizing over the global policy means optimizing over its parameters ϕ , since $\pi_G = \pi_\phi$. A saddle point of (14) is approached by ascending $\mathcal{L}_G$ in ϕ and descending it in μ . Both directions need $\nabla_\phi \mathcal{L}_G$, built from the per-task gradients $\nabla_\phi V_i^{\pi_\phi}$. The derivation (15)–(17) used nothing about θ beyond the fact that only the policy factors of the trajectory law (3) carry the parameter, so it holds verbatim with ϕ in place of θ ,

$$\nabla_\phi V_i^{\pi_\phi} = \mathbb{E}_{\tau \sim p_i^{\pi_\phi}} \left[G_i(\tau) \sum_{t=0}^T \nabla_\phi \log \pi_\phi(a_t | s_t) \right]. \quad (25)$$

During the global update, the local values $V_i^{\pi_L}$ and $V_k^{\pi_L}$ are fixed with respect to ϕ . For the squared shortfall in (11),

$$\nabla_\phi F_G(\pi_\phi) = \begin{cases} -2(V_k^{\pi_L} - V_k^{\pi_\phi}) \nabla_\phi V_k^{\pi_\phi}, & \text{if } V_k^{\pi_\phi} < V_k^{\pi_L}, \\ 0, & \text{if } V_k^{\pi_\phi} \geq V_k^{\pi_L}. \end{cases} \quad (26)$$

Differentiating (13) then gives

$$\nabla_\phi \mathcal{L}_G(\pi_\phi, \mu) = \sum_{i=1}^{k-1} \omega_i \nabla_\phi V_i^{\pi_\phi} + \mu \begin{cases} 2(V_k^{\pi_L} - V_k^{\pi_\phi}) \nabla_\phi V_k^{\pi_\phi}, & \text{if } V_k^{\pi_\phi} < V_k^{\pi_L}, \\ 0, & \text{if } V_k^{\pi_\phi} \geq V_k^{\pi_L}, \end{cases} \quad (27)$$

each per-task gradient given by (25).

Drawing N trajectories $\tau^{(1)}, \dots, \tau^{(N)}$ on task i under π_ϕ , the per-task value and gradient are estimated by

$$\hat{V}_i^{\pi_\phi} = \frac{1}{N} \sum_{n=1}^N G_i(\tau^{(n)}), \quad (28)$$

and

$$\overline{\nabla_\phi V_i^{\pi_\phi}} = \frac{1}{N} \sum_{n=1}^N G_i(\tau^{(n)}) \sum_{t=0}^T \nabla_\phi \log \pi_\phi(a_t^{(n)} | s_t^{(n)}). \quad (29)$$

During the same global update, $\hat{V}_k^{\pi_L}$ denotes the rollout estimate of the held-fixed local value on the current task. The global primal update is

$$\phi^{(m+1)} = \phi^{(m)} + \beta \left[\sum_{i=1}^{k-1} \omega_i \overline{\nabla_\phi V_i^{\pi_\phi}} + \mu^{(m)} \begin{cases} 2(\hat{V}_k^{\pi_L} - \hat{V}_k^{\pi_\phi}) \overline{\nabla_\phi V_k^{\pi_\phi}}, & \text{if } \hat{V}_k^{\pi_\phi} < \hat{V}_k^{\pi_L}, \\ 0, & \text{if } \hat{V}_k^{\pi_\phi} \geq \hat{V}_k^{\pi_L}. \end{cases} \right]. \quad (30)$$

The global multiplier update is

$$\mu^{(m+1)} = \max \left\{ 0, \mu^{(m)} + \eta_\mu (\hat{F}_G^{(m)} - \epsilon) \right\}, \quad (31)$$

where

$$\hat{F}_G^{(m)} = \begin{cases} (\hat{V}_k^{\pi_L} - \hat{V}_k^{\pi_\phi^{(m)}})^2, & \text{if } \hat{V}_k^{\pi_\phi^{(m)}} < \hat{V}_k^{\pi_L}, \\ 0, & \text{if } \hat{V}_k^{\pi_\phi^{(m)}} \geq \hat{V}_k^{\pi_L}. \end{cases} \quad (32)$$

After the global phase for task k , the new task becomes part of the past-task set for the next task.